Kulasekaran M

+91 959-730-4897 | kulasekarankr1@gmail.com | linkedin.com/in/kulasekaran-m | Kulasekaran M Portfolio

Professional Summary

Embedded Firmware Engineer with expertise in STM32, ESP32, Raspberry Pi, and RTOS development. Proven experience implementing real-time computer vision systems, IoT protocols (MQTT, HTTP/HTTPS), and AWS cloud integration. Proficient in C, Python, hardware debugging , and communication protocols (SPI, I2C, UART). Strong background in ARM architecture, embedded systems optimization, and machine learning integration for IoT applications.

Technical Skills

Programming Languages: C, Python, Assembly Language,VHDL
Embedded Systems: ESP32, Raspberry Pi, Jetson AGX Xavier, ARM Cortex-M,FPGA
RTOS & Protocols: FreeRTOS, SPI, I2C, UART, MQTT, HTTP/HTTPS, MAVLink
Development Tools: STM32CubeIDE, Keil µVision, GDB, Git, Oscilloscope, Logic Analyzer
Cloud & AI/ML: AWS (S3, Lambda, IoT Core), YOLOv5, OpenCV, Computer Vision, Machine Learning
Hardware Design: PCB Design (KiCad), Cadence, Circuit Debugging, Power Management

Professional Experience

R&D Embedded Engineer June 2024 – July 2025
BB IoT Solutions Pvt Ltd *Vellore, Tamil Nadu*

- Developed real-time object recognition system integrating computer vision and RFID technology, achieving **68% to 92% accuracy YOLO + RFID fusion** for industrial IoT applications
- Maintained **99.5% uptime** uptime over 6 months in production handling 15K+ MQTT messages/day in production environment
- Collaborated with cross-functional teams to optimize firmware for memory efficiency and power consumption in resource-constrained embedded devices

Publications

Advanced Edge-Cloud Offloading System for Industrial Quality Control Yet to be Published
Research Paper – Real-Time Embedded Systems

- Designed RTOS-based embedded system with intelligent edge-cloud workload distribution for post-manufacturing chip defect detection under CPU, thermal, and power constraints
- Achieved **30% reduction in edge CPU utilization** through dynamic cloud offloading while maintaining real-time performance requirements using MQTT/HTTP communication

Technical Projects

Embedded mmWave Radar Signal Acquisition and Motion Detection System using ESP32-S3 | SPI, FMCW Radar

- Interfaced Infineon BGT60TR13C 60GHz FMCW radar sensor with ESP32-S3 using high-speed SPI for real-time data acquisition
- Configured radar registers and implemented firmware pipeline to read raw ADC data from on-chip FIFO
- Developed lightweight signal processing (energy-based detection) to enable motion and presence detection under resource constraints
- Optimized data handling and communication to manage high-throughput radar streams while maintaining real-time system responsiveness

Edge AI Enabled Autonomous Disaster Response Drone With Multi-sensor Perception | **Published**

- Built autonomous disaster response drone with S500 quadcopter frame, Pixhawk flight controller, and multi-sensor payload (RGB/thermal cameras, gas sensors MQ2/MQ7)
- Integrated sensor suite using I2C and SPI protocols with real-time telemetry via MAVLink communication protocol
- Implemented YOLOv5 object detection on Raspberry Pi for real-time target identification, achieving **85% accuracy** in disaster simulation scenarios

Secure Embedded Linux Multi-Sensor Surveillance System with Kernel Drivers and Firmware Integrity Protection | *I2C, V4L2*

- Designed and implemented custom Linux kernel I2C device drivers for BME280 and APDS9960 sensors with interrupt-driven detection and character device interfaces
- Developed real-time video/audio capture system using V4L2 and ALSA with memory-mapped buffer management, achieving sub-50ms event-triggered recording latency
- Implemented firmware integrity verification using periodic cryptographic hashing to detect tampering and prevent malicious firmware modification
- Designed lightweight security mechanism to restrict unauthorized updates and ensure reliable operation under potential physical and software-level attacks.

Hardware-Accelerated Multi-Stage Appliance Control System Using FPGA and VHDL | *VHDL, FSM, Xilinx Vivado, FPGA*

- Designed and implemented a Finite State Machine (FSM)-based control system for automating washing machine operations including water fill, wash, rinse, drain, and spin cycles
- Developed VHDL behavioral code for state transitions, timing control, and output logic using synchronous design principles
- Achieved reliable and efficient hardware-level automation, improving flexibility and performance over traditional microcontroller-based designs

Education

Vellore Institute of Technology — Vellore, Tamil Nadu
Master of Technology (M.Tech) in Embedded Systems; CGPA: 7.69/10 — *Pursuing*

Vellore Institute of Technology — chennai, Tamil Nadu
Bachelor of Technology (B.Tech) in Electronics and Communication Engineering; CGPA: 7.88/10 — *2020 – 2024*

Certifications & Professional Development

AWS Certification – AWS Certified Cloud Practitioner (2024)

Berribot Pvt Ltd [Internship] – AI Testing Engineer (May – July 2024)

PCB Design Using KiCad – KiCad Workshop Certification

Design of High-Speed VLSI Circuits Using Cadence – VLSI Cadence

24hrs Industrial Conclave Hack-a-thon – Winner

8-Hours NetSim Hackathon 2.0 – Participant

Leadership & Extracurricular Activities

IEEE Student Chapter, VIT — 2020 – 2024
Student Coordinator — *chennai, Tamil Nadu*

* Coordinated 12+ technical sessions and guest lectures.
* Organized embedded systems workshop with 150+ participants.
* Increased membership engagement by 35%.

Wayspire EdTech — 2023 – 2024
College Ambassador

* Acted as liaison between organization and student body, increasing internship participation by **20%**